OR-AS Operations Research Applications and Solutions	Case Name: **CRM System**	Sector	**IT**		
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	■
			Schedule with costs	■
		Risk Analysis	Random simulation	■
			One of nine std. scenarios	
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions	
Date	**2015**	Project Control	Automatic tracking	
File Name	**C2015-24 CRM System**		Tracking based on user input	■

1. Project description

Project authenticity

A CRM system project involving requirements gathering, solution analysis, data and channel design, development, UAT testing, and post-go-live support.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	29
Planned Duration (PD)	233 days*
Budget At Completion (BAC)	€ 44,130
Renewable Resources	12
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	7%
Activity Distribution (AD)	59%
Length of Arcs (LA)	7%
Topological Float (TF)	29%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	9.1	16.7	-4.2
CRI-rho	23.2	23.0	-1.2
CRI-tau	37.9	41.2	-0.8

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	8.5	13.5	-5.8
CRI-rho	8.5	13.5	-5.8
CRI-tau	8.5	13.5	-5.8

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	7.1	25.3	-3.6
SI	10.2	19.5	-3.8
SSI	3.4	18.2	-5.4
CRI-r	10.2	18.7	-4.2
CRI-rho	24.0	24.0	-1.3
CRI-tau	40.8	41.0	-0.8

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	92.4	-35.7
PV - SPI	60.8	60.8
PV - SCI	61.8	71.8
ED - 1	1310.0	1307.7
ED - SPI	60.8	60.8
ED - SCI	61.3	61.3
ES - 1	142.3	-142.1
ES - SPI(t)	22.6	3.9
ES - SCI(t)	22.8	3.8

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	2.3	0.6
CPI	1.8	0.7
SPI	10.5	10.5
SPI(t)	10.1	10.1
SCI	10.5	10.5
SCI(t)	10.1	10.1
0.8 CPI + 0.2 SPI	9.0	9.0
0.8 CPI + 0.2 SPI(t)	8.0	8.0

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 6 tracking periods with a length of approximately 2 months. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	4875.1	-3987.4	-71.7	1.4	1.0	1.1	0.8
std dev	2868.8	4898.8	215.2	0.1	0.6	0.5	0.2
final	7260.0	1050.0	7.0	1.2	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	233 days	Real Duration	233 days	On Time	0%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	225.3	1.6	0.5	0.5
PV - SPI	226.0	1.3	0.5	0.5
PV - SCI	226.7	1.0	0.5	0.5
ED - 1	227.3	1.0	0.4	0.4
ED - SPI	228.0	1.3	0.4	0.4
ED - SCI	228.7	1.6	0.3	0.3
ES - 1	229.5	1.9	0.3	0.3
ES - SPI(t)	230.3	1.6	0.2	0.2
ES - SCI(t)	231.0	1.3	0.1	0.1

2.3.3.3. Cost forecasting

BAC	€ 44,130	Real Cost	€ 36,870	Under Budget	-16.5%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	39254.9	2868.8	1.1	-1.1
CPI	32828.8	3214.9	1.8	1.8
SPI	43905.2	17048.3	3.2	-3.2
SPI(t)	39568.0	11645.2	1.2	-1.2
SCI	36073.8	11629.5	0.4	0.4
SCI(t)	33074.8	9234.2	1.7	1.7
0.8 CPI + 0.2 SPI	33599.0	3460.1	1.5	1.5
0.8 CPI + 0.2 SPI(t)	33404.1	3340.0	1.6	1.6